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Knee joint

Course notes intended for first-year medical students.

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Knee joint

Course outline

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1. Articular surfaces
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Knee joint

Introduction

The knee joint is a large, superficial joint that connects the thigh and the lower leg. It is a weight-bearing joint,reinforced with ligaments and tendons.
This arrangement provides the stability necessary for standing (on two legs) and walking (on one leg).
The knee joint is composed of two anatomically and functionally inseparable articulations, enclosed within the same joint capsule:

- The patellofemoral joint, or ginglyme, is the primary element of the extensor mechanism of the knee.
- The bicondylar femorotibial joint is involved in flexion-extension movements.

1.Articular surfaces

1.1.The patellar surface of the femur

It is located on the anterior face of the distal epiphysis of the femur, and it articulates with the posterior face of the patella. Its lateral slope is more extensive than its medial slope.

1.2.The femoral condyles

There are two of them, one medial and the other lateral; they are curved articular surfaces rolled upon themselves.
They reply:

- Below: at the menisci and the 2 superior tibial articular surfaces.
- Anteriorly: on the posterior side of the patella.

1.3.The tibial plateau

The tibial plateau comprises the superior tibial articular surfaces (glenoid cavities) that articulate with the menisci and condyles of the femur. The two articular surfaces are separated by the intercondylar space, which is occupied by the intercondylar tubercles (tibial spines).

1.4.The posterior surface of the patella

It is articular in its upper 2/3, it corresponds to the patellar surface of the femur.

1.5.The menisci of the knee

The menisci are intra-articular, semi-lunar fibrocartilages interposed between the femoral condyles and the superior tibial articular surfaces, whose role is to increase
The concavity of the superior tibial articular surfaces (tibial glenoids). There are two of them :

- the medial meniscus, shaped like a "C".
- The lateral meniscus, shaped like an "O".

2. Means of union

The following structures maintain contact between the joint surfaces: the joint capsule, passive ligaments, and active ligaments.

2.1.The joint capsule

The joint capsule is a fibrous sleeve that connects the articular surfaces. It inserts close to the articular cartilage and has the shape of a cylinder.
At the front, it presents a window in which the patella is placed.
Laterally it adheres to the peripheral face of the menisci.
Posteriorly it thickens opposite the femoral condyles where it forms the condylar shells. The lateral shell contains an inconstant sesamoid bone: the fabula.

2.2.Passive ligaments

Ligaments provide stability to the knee, which is superficial and exposed to trauma, particularly in athletes; these ligaments form 3 systems:

2.2.1. The collateral system

It ensures the collateral stability of the knee, it is represented by: the fibular and tibial collateral ligaments.

ro gnippag sa hcus ,tnemevom lamronba ni stluser metsys tnemagil laretalloc eht ot yrujnI .devresbo yllacinilc si hcihw ,ytixal laretal ro laidem

The tibial collateral ligament (medial collateral ligament) is epiphyseal–diaphyseal.

.laesyhpaid–laesyhpipe si (tnemagil laretalloc laretal) tnemagil laretalloc ralubif ehT

2.2.2. The central pivot

The central pivot ensures the anteroposterior stability of the knee; it is represented by the cruciate ligaments.

The anterolateral cruciate ligament extends from the anterior intercondylar area. It runs posteriorly and laterally. It inserts on the medial surface of the lateral condyle.

The posteromedial cruciate ligament extends from the posterior intercondylar area. Its course is vertical, ending on the lateral aspect of the medial condyle.

A cruciate ligament injury results in the drawer sign, which is clinically assessed.

2.2.3. The sagittal system

The sagittal system is represented by:

- Anteriorly, the patellar ligament, inseparable from the quadriceps tendon, extends from the patellar apex to the anterior tibial tuberosity.
- Posteriorly the arcuate ligament extends from the fibular head in a fan shape to terminate on the lateral condylar shell and the fabula and on the medial condylar shell.

  Noticed :

- Patellar retinaculum (patellar wings): connect the patella to each epicondyle; the medial one is wider and stronger than the lateral one.
- Menisco-patellar ligaments connect the patella to the menisci.

2.3. Active Ligaments

It is the muscles that contribute to joint stability through their tendon endings.

- **Ahead**

The quadriceps tendon continues as the patellar ligament. This forms the extensor system of the knee, composed of the quadriceps muscle, quadriceps tendon, patella, and patellar tendon.

- **Back**

The semimembranosus muscle, the biceps femoris muscle and the popliteus muscle.

- **Medically**

The muscles of the "goose feet" (sartorius, gracilis and semitendinosus).

- **Laterally**

Expansion of the tensor fasciae latae muscle.

3. Synovial

It lines the deep surface of the joint capsule, and in front it forms a large cul-de-sac, which is the quadriceps cul-de-sac.

4. Joint physiology

The knee joint allows the following movements:

- Flexion-extension movements
- Medial rotation-lateral rotation movements

4.1. Flexion-extension movements

The reference position: the axis of the leg is in line with the axis of the thigh.
Extension is the act of bringing the front of the leg closer to the front of the thigh.

- The extension is locked; it only exists for a specific flexed position.

Flexion is bringing the back of the leg closer to the back of the thigh.

- An active flexion (muscle action) with a range of motion of 140° for a flexed hip, and with a range of motion of 120° for an extended hip.
- Passive flexion: the subject sitting on their heels at 160° amplitude (squatting station or squatting position).

4.2. Medial rotation – lateral rotation movements
The reference position: knee bent at 90°, toes pointed forward.

- Medial rotation brings the toes inward at 30°.

- Lateral rotation brings the toes outwards at 40°.

5. Clinical application

5.1. Radioanatomy

Numerous techniques are available today for analyzing the tendons, bone, capsuloligamentous structures, menisci, trophicity, and muscular content of the knee joint. These techniques use ultrasound, computed tomography (CT) scans, with or without arthrography, and magnetic resonance imaging (MRI), with or without arthrography.

5.2. Pathologies

The knee joint is a site of trauma, especially among athletes; the most frequent injury is a meniscal tear.

Osteoarthritis (gonarthrosis) is a common condition in older people.

Conclusion

The knee joint is a large, superficial joint that connects the thigh and the lower leg. It is a weight-bearing joint, equipped with ligaments and tendons.

This device ensures the stability necessary for standing (in bipedal support) and during walking (in unipedal support).

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